

Synthetic Hospital Knowledge Base for RAG-Based Indoor Navigation & Doctor Finder

Final Year Project – Prototype Dataset / Demonstration Scenario

Purpose: This document contains a fully synthetic hospital scenario designed for a Retrieval-Augmented Generation (RAG) application. The chatbot should use this knowledge base to answer questions such as “Where is the dental block?”, “I have severe tooth pain—which doctor should I see?”, “Which floor is the eye department on?”, and “Which doctor is available after 4 PM?”. All names, rooms, timings, and locations below are fictional and intended only for project demonstration.

1. Hospital Layout Overview

Assume a large government hospital campus with five major buildings (Block 1–Block 5). Every block has exactly three floors: Ground Floor, First Floor, and Second Floor. Block 1 is dedicated to pharmacy and medicine-related services. Blocks 2–5 contain clinical departments, diagnostic services, specialist clinics, emergency-related services, and supporting facilities.

Block	Primary Function	Floors	Key Services
Block 1	Pharmacy & Medicine Services	Ground, First, Second	Main Pharmacy, Dispensary, Medical Store
Block 2	Dental & ENT Services	Ground, First, Second	Dental OP, Oral Surgery, ENT
Block 3	Eye, General Medicine & Cardiology	Ground, First, Second	Eye, General Medicine, Cardiology
Block 4	Orthopaedics, Neurology & Physiotherapy	Ground, First, Second	Bone/Joint, Neurology, Rehab
Block 5	Women & Child Health	Ground, First, Second	Obstetrics, Gynaecology, Paediatrics

2. Block and Floor Directory

Block	Floor	Departments / Services	Location Hint
Block 1	Ground Floor	Main Pharmacy; Prescription Counter; Token Counter	Near Main Gate; beside Registration
Block 1	First Floor	Inpatient Pharmacy; Drug Information Desk; IV/Injection Store	Above Ground Pharmacy
Block 1	Second Floor	Medical Store; Bulk Medicine Distribution; Administrative Pharmacy Office	Restricted service area
Block 2	Ground Floor	Dental OP Registration; General Dental Clinic; Dental Waiting Area	East side of campus
Block 2	First Floor	Oral & Maxillofacial Surgery; Prosthodontics; Dental X-Ray	Lift/Staircase from Dental OP
Block 2	Second Floor	ENT OP; Audiology; ENT Procedure Room	Above Dental First Floor
Block 3	Ground Floor	General Medicine OP; Vital Signs Desk; General Consultation Rooms	Central clinical zone
Block 3	First Floor	Ophthalmology/Eye Clinic; Eye Testing; Retina Clinic	Near central corridor
Block 3	Second Floor	Cardiology; ECG; Echo; Cardiac Consultation	North side of Block 3
Block 4	Ground Floor	Orthopaedics OP; Fracture Clinic; Plaster Room	West side of campus
Block 4	First Floor	Neurology; Neurosurgery Consultation; EEG	Above Orthopaedics

Block 4	Second Floor	Physiotherapy; Rehabilitation; Mobility Training	Near rear access road
Block 5	Ground Floor	Paediatrics OP; Vaccination; Child Emergency Consultation	South side of campus
Block 5	First Floor	Obstetrics & Gynaecology OP; Antenatal Clinic	Women & Child Health zone
Block 5	Second Floor	Maternity Ward; Postnatal Consultation; Lactation Support	Restricted ward floor

3. Doctor Directory and Availability

Specialty	Block	Floor	Room	Doctor	Specialization	Timing	Days
Dental	Block 2	Ground	Room D-101	Dr. Meena Krishnan	General Dentist	09:00–16:00	Mon–Sat
Dental	Block 2	Ground	Room D-102	Dr. Arjun Kumar	General & Emergency Dentist	16:00–22:00	Mon–Sat
Dental	Block 2	First	Room D-201	Dr. Nivetha Raman	Oral & Maxillofacial Surgeon	10:00–15:00	Mon–Fri
Dental	Block 2	First	Room D-202	Dr. Karthik Raj	Prosthodontist	09:00–14:00	Mon–Sat
ENT	Block 2	Second	Room E-201	Dr. Priya Menon	ENT Specialist	09:00–15:00	Mon–Sat
ENT	Block 2	Second	Room E-202	Dr. Rahul Dev	ENT Specialist	15:00–21:00	Mon–Sat
General Medicine	Block 3	Ground	Room GM-101	Dr. Suresh Babu	General Physician	09:00–16:00	Mon–Sat
General Medicine	Block 3	Ground	Room GM-102	Dr. Asha Devi	General Physician	16:00–22:00	Mon–Sat
Ophthalmology	Block 3	First	Room OPH-101	Dr. Lakshmi Narayan	Ophthalmologist	09:00–16:00	Mon–Sat
Ophthalmology	Block 3	First	Room OPH-102	Dr. Vivek Anand	Ophthalmologist	16:00–21:00	Mon–Sat
Cardiology	Block 3	Second	Room CAR-201	Dr. Hari Prasad	Cardiologist	09:00–15:00	Mon–Fri
Cardiology	Block 3	Second	Room CAR-202	Dr. Sneha Iyer	Cardiologist	15:00–21:00	Mon–Fri
Orthopaedics	Block 4	Ground	Room ORT-101	Dr. Manoj Kumar	Orthopaedic Surgeon	09:00–16:00	Mon–Sat
Orthopaedics	Block 4	Ground	Room ORT-102	Dr. Deepa Shah	Orthopaedic Specialist	16:00–22:00	Mon–Sat
Neurology	Block 4	First	Room NEU-201	Dr. Vinoth Kumar	Neurologist	09:00–14:00	Mon–Fri
Neurology	Block 4	First	Room NEU-202	Dr. Anjali Rao	Neurologist	14:00–20:00	Mon–Fri
Physiotherapy	Block 4	Second	Room PHY-301	Dr. Farah Ali	Physiotherapist	09:00–15:00	Mon–Sat
Physiotherapy	Block 4	Second	Room PHY-302	Dr. Naveen Kumar	Physiotherapist	15:00–21:00	Mon–Sat
Paediatrics	Block 5	Ground	Room PED-101	Dr. Kavitha Rao	Paediatrician	09:00–16:00	Mon–Sat
Paediatrics	Block 5	Ground	Room PED-102	Dr. Imran Sheikh	Paediatrician	16:00–22:00	Mon–Sat
Obstetrics	Block 5	First	Room OBG-201	Dr. Revathi Menon	Obstetrician	09:00–16:00	Mon–Sat
Gynaecology	Block 5	First	Room GYN-202	Dr. Swetha R	Gynaecologist	16:00–22:00	Mon–Sat
Maternity	Block 5	Second	Room MAT-301	Dr. Priyanka Das	Obstetrician	09:00–15:00	Mon–Sat
Maternity	Block 5	Second	Room MAT-302	Dr. Joseph Mathew	Obstetrician	15:00–21:00	Mon–Sat

4. Pharmacy Services

Service	Block	Floor	Room	Timing	Description
Main Pharmacy	Block 1	Ground	P-101	08:00–22:00	Prescription medicines and routine dispensing
Emergency Pharmacy Counter	Block 1	Ground	P-102	24 hours	Emergency medicines; subject to hospital policy
Inpatient Pharmacy	Block 1	First	P-201	08:00–20:00	Medicines for admitted patients
Drug Information Desk	Block 1	First	P-202	09:00–17:00	Medicine-related information
Medical Store	Block 1	Second	P-301	08:00–18:00	Bulk stock; not general public counter

5. Symptom-to-Department Knowledge

User Symptom / Request	Department	Block	Floor	Room	Doctor Type	RAG Routing Note
Severe tooth pain	Dental	Block 2	Ground Floor	D-101 / D-102	General Dentist / Emergency Dentist	If after 16:00, recommend Dr. Arjun Kumar
Swollen gums	Dental	Block 2	Ground Floor	D-101 / D-102	General Dentist / Emergency Dentist	Dental evaluation
Broken or impacted tooth	Dental / Oral Surgery	Block 2	First Floor	D-201	Oral & Maxillofacial Surgeon	Recommend oral surgery consultation
Need dentures / replacement teeth	Dental	Block 2	First Floor	D-202	Prosthodontist	Prosthodontic consultation
Ear pain	ENT	Block 2	Second Floor	E-201 / E-202	ENT Specialist	Choose based on current time
Hearing difficulty	ENT	Block 2	Second Floor	E-201 / Audiology	ENT Specialist / Audiology	Audiology testing available
Eye pain / blurred vision	Ophthalmology	Block 3	First Floor	OPH-101 / OPH-102	Ophthalmologist	Choose available doctor
Chest pain	Cardiology / Emergency	Block 3	Second Floor	CAR-201 / CAR-202	Cardiologist	For severe/emergency symptoms, direct to emergency services rather than routine OP
Fever / general illness	General Medicine	Block 3	Ground Floor	GM-101 / GM-102	General Physician	Choose available doctor
Knee pain	Orthopaedics	Block 4	Ground Floor	ORT-101 / ORT-102	Orthopaedic Specialist	Physiotherapy may be recommended after evaluation
Back pain	Orthopaedics	Block 4	Ground Floor	ORT-101 / ORT-102	Orthopaedic Specialist	Choose available doctor
Headache with neurological symptoms	Neurology	Block 4	First Floor	NEU-201 / NEU-202	Neurologist	Urgent symptoms should be escalated to emergency care
Child fever	Paediatrics	Block 5	Ground Floor	PED-101 / PED-102	Paediatrician	For children
Pregnancy	Obstetrics	Block 5	First Floor	OBG-201	Obstetrician	Antenatal clinic

consultation						
Women's reproductive health issue	Gynaecology	Block 5	First Floor	GYN-202	Gynaecologist	Routine consultation

6. Expected RAG Chatbot Responses

User: “Where is the dental block?”

Dental services are in Block 2. General Dental OP is on the Ground Floor, Rooms D-101 and D-102. Block 2 is on the east side of the hospital campus. From the main gate, follow the main corridor towards the east clinical zone and look for the Block 2 / Dental signage.

User: “I have severe tooth pain. Where should I go?”

You should visit the Dental Department in Block 2, Ground Floor. Dr. Meena Krishnan (General Dentist) is available from 9:00 AM–4:00 PM in Room D-101. From 4:00 PM–10:00 PM, Dr. Arjun Kumar (General & Emergency Dentist) is available in Room D-102. If the current time is after 4 PM, the chatbot should prefer Dr. Arjun Kumar.

User: “Who is the dental doctor after 4 PM?”

Dr. Arjun Kumar, General & Emergency Dentist, is available in Block 2, Ground Floor, Room D-102, from 4:00 PM–10:00 PM, Monday–Saturday.

User: “Where is the eye department?”

The Ophthalmology Department is in Block 3, First Floor. Dr. Lakshmi Narayan is available 9:00 AM–4:00 PM in Room OPH-101, and Dr. Vivek Anand is available 4:00 PM–9:00 PM in Room OPH-102.

User: “I have knee pain. Which doctor should I see?”

The Orthopaedics Department is in Block 4, Ground Floor. Dr. Manoj Kumar is available 9:00 AM–4:00 PM in Room ORT-101. Dr. Deepa Shah is available 4:00 PM–10:00 PM in Room ORT-102.

User: “Where can I get medicine after seeing the doctor?”

The Main Pharmacy is in Block 1, Ground Floor, Room P-101, and is open 8:00 AM–10:00 PM. The Emergency Pharmacy Counter is in Block 1, Ground Floor, Room P-102 and operates 24 hours.

7. Navigation and Nearby-Location Metadata

Landmark / Destination	Purpose	Block	Nearby / Direction Hint
Main Gate	Starting landmark	Block 1	Block 1 is closest to the Main Gate
Registration Counter	Registration	Block 1	Near Block 1 Ground Floor
Block 1 Pharmacy	Pharmacy	Block 1	Ground Floor
Block 2 Dental	Dental	Block 2	East clinical zone; next to the central corridor
Block 2 ENT	ENT	Block 2	Second Floor
Block 3 General Medicine	General Medicine	Block 3	Central clinical zone
Block 3 Eye	Ophthalmology	Block 3	First Floor
Block 3 Cardiology	Cardiology	Block 3	Second Floor
Block 4 Orthopaedics	Orthopaedics	Block 4	West clinical zone
Block 4 Neurology	Neurology	Block 4	First Floor
Block 4 Physiotherapy	Physiotherapy	Block 4	Second Floor
Block 5 Paediatrics	Paediatrics	Block 5	South clinical zone
Block 5 Obstetrics	Obstetrics	Block 5	First Floor
Block 5 Maternity	Maternity	Block 5	Second Floor

8. RAG-Friendly Facts / Retrieval Chunks

- FACT: Dental Department = Block 2. Dental General OP = Ground Floor. Rooms D-101 and D-102.

- FACT: Dental doctor Dr. Meena Krishnan = General Dentist = Block 2 Ground Floor Room D-101 = 09:00–16:00 Mon–Sat.
- FACT: Dental doctor Dr. Arjun Kumar = General & Emergency Dentist = Block 2 Ground Floor Room D-102 = 16:00–22:00 Mon–Sat.
- FACT: Oral & Maxillofacial Surgery = Block 2 First Floor Room D-201 = Dr. Nivetha Raman = 10:00–15:00 Mon–Fri.
- FACT: Prosthodontics = Block 2 First Floor Room D-202 = Dr. Karthik Raj = 09:00–14:00 Mon–Sat.
- FACT: ENT = Block 2 Second Floor. Dr. Priya Menon 09:00–15:00; Dr. Rahul Dev 15:00–21:00.
- FACT: Ophthalmology/Eye Clinic = Block 3 First Floor. Dr. Lakshmi Narayan 09:00–16:00; Dr. Vivek Anand 16:00–21:00.
- FACT: General Medicine = Block 3 Ground Floor. Dr. Suresh Babu 09:00–16:00; Dr. Asha Devi 16:00–22:00.
- FACT: Cardiology = Block 3 Second Floor. Dr. Hari Prasad 09:00–15:00; Dr. Sneha Iyer 15:00–21:00.
- FACT: Orthopaedics = Block 4 Ground Floor. Dr. Manoj Kumar 09:00–16:00; Dr. Deepa Shah 16:00–22:00.
- FACT: Neurology = Block 4 First Floor. Dr. Vinoth Kumar 09:00–14:00; Dr. Anjali Rao 14:00–20:00.
- FACT: Physiotherapy = Block 4 Second Floor. Dr. Farah Ali 09:00–15:00; Dr. Naveen Kumar 15:00–21:00.
- FACT: Paediatrics = Block 5 Ground Floor. Dr. Kavitha Rao 09:00–16:00; Dr. Imran Sheikh 16:00–22:00.
- FACT: Obstetrics = Block 5 First Floor. Dr. Revathi Menon 09:00–16:00.
- FACT: Gynaecology = Block 5 First Floor. Dr. Swetha R 16:00–22:00.
- FACT: Maternity = Block 5 Second Floor. Dr. Priyanka Das 09:00–15:00; Dr. Joseph Mathew 15:00–21:00.
- FACT: Main Pharmacy = Block 1 Ground Floor Room P-101 = 08:00–22:00.
- FACT: Emergency Pharmacy Counter = Block 1 Ground Floor Room P-102 = 24 hours.
- FACT: Every block has three floors: Ground Floor, First Floor, Second Floor.
- FACT: Block 1 is the Pharmacy block; Blocks 2–5 contain clinical departments.

9. Recommended RAG Answer Logic

1. Identify the user's intent: location, department, symptom, doctor, room, floor, pharmacy, or timing.
2. Retrieve the relevant block/floor/room facts from the knowledge base.
3. If a symptom is given, map the symptom to the most relevant department using the symptom-to-department table.
4. If the user asks for a doctor, filter doctors by department and current day/time before recommending one.

5. If two doctors cover different time windows, recommend the currently available doctor and mention the alternate doctor's timing.
 6. Always provide Block + Floor + Room when a specific destination is known.
 7. Add a short nearby landmark/direction hint when answering location questions.
 8. If the user asks "where is X?", do not answer only with the department name; include block and floor.
 9. If the requested department is not present in the dataset, state that it is not available in the current hospital knowledge base instead of inventing a location.
 10. This dataset is synthetic. For a real hospital deployment, doctor availability, room numbers, emergency routing, and timings must be connected to an authoritative hospital database and updated in real time.
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